ECE 3337 — Mock Exam

Topics: Signals & Systems + Fourier Series/Transforms + Laplace Transforms | Total: 80 points | Time: 90 minutes

1. **(5 points)** Starting with Euler's relationship $e^{j\theta} = \cos\theta + j \sin\theta$, show that:

$$\sin(2x) = 2 \sin x \cos x$$

2. **(5 points)** Starting from the definition of the Dirac delta function, derive the sifting property $\int f(t)\delta(t - t_0)\, dt = f(t_0)$. Then use it to calculate:

$$y(t) = \int_{-5}^{-2} t^4\delta(t + 3)\, dt + \int_{1}^{6} t^4\delta(t - 8)\, dt$$

3. **(10 points)** Starting with the original sine/cosine Fourier representation,

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

derive the exponential form of the Fourier series, $x(t) = \sum_{n=-\infty}^{\infty} x_n e^{jn\omega_0 t}$. For full credit, you must also derive the expression for x_n as an integral over one period.

4. **(5 points)** Using the result from Question 3, derive Parseval's theorem for the exponential form of the Fourier series.

5. **(10 points)** Compute the exponential Fourier series coefficients x_n of an impulse train:

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

Then use the computed series to find the Fourier transform $X(\omega)$ of the impulse train. What can you conclude about the spectrum of a periodic impulse train from your result?

6. **(5 points)** Show that the Fourier transform of an even, real-valued signal $x(t)$ is given by:

$$X(\omega) = 2\int_{0}^{\infty} x(t)\cos(\omega t)\, dt$$

7. **(10 points)** Starting from the definition $X(s) = \int_0^{\infty} x(t)e^{-st}\, dt$, derive the Laplace transform of:

$$x(t) = e^{-at}\cos(\omega_0 t)\, u(t), \quad a > 0$$

Then explain how the Fourier transform of the same signal can be obtained from your Laplace transform result, and state the condition on a that makes this substitution valid.

8. **(10 points)** Define a bounded signal, then derive the necessary and sufficient condition for an LTI system with impulse response $h(t)$ to be BIBO stable. Use the condition you

derived to determine whether the system below is BIBO stable:

$$h(t) = e^{-2t}\cos(3t)\, u(t)$$

9. **(5 points)** Derive the Laplace transform convolution property, ■$\{x_1(t) * x_2(t)\} = X_1(s)X_2(s)$, starting from the definition of convolution and swapping the order of integration. State clearly where you use the fact that both signals are causal.

10. **(10 points)** Derive the Laplace transform of $x'(t)$ and $x''(t)$ in terms of $X(s)$, $x(0^-)$, and $x'(0^-)$. Then use your results to solve the following differential equation for $y(t)$, $t \geq 0$, given $y(0^-) = 2$ and $y'(0^-) = -1$:

$$y''(t) + 3y'(t) + 2y(t) = 0$$